Logistic Regression

Logistic Regression is a probabilistic model used to establish the relationship between a binary dependent variable Y and one or more independent variables X_i. The model predicts the probability of a binary outcome, and the output lies between 0 and 1.

Let $p = P(Y = 1|X)$. The logistic regression model is defined as:

$$\ln\left(\frac{p}{1-p}\right) = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

Here,

$$\ln\left(\frac{p}{1-p}\right)$$

is called the **log-odds (logit)** function.

From Log-Odds to Probability

Exponentiating both sides:

$$\frac{p}{1-p} = e^{b_0+b_1x_1+\cdots+b_nx_n}$$

Solving for p:

$$p = \frac{e^{b_0+b_1x_1+\cdots+b_nx_n}}{1 + e^{b_0+b_1x_1+\cdots+b_nx_n}}$$

This can be written as:

$$p = \frac{1}{1 + e^{-(b_0+b_1x_1+\cdots+b_nx_n)}}$$

Sigmoid Function

The above expression is the **Sigmoid Function**:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Derivative of Sigmoid Function

$$\sigma(x) = (1 + e^{-x})^{-1}$$

Differentiating:

$$\sigma'(x) = -(1 + e^{-x})^{-2} \cdot (-e^{-x})$$

$$\sigma'(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$$

Rewriting:

$$\sigma'(x) = \sigma(x)\,(1 - \sigma(x))$$

Bernoulli Assumption

Since $Y \in \{0, 1\}$, we assume:

$$Y \sim \text{Bernoulli}(p)$$

Bernoulli probability mass function:

$$P(Y = y) = p^y (1 - p)^{1-y}$$

Binary Cross-Entropy Loss

For m independent samples, the likelihood is:

$$\mathcal{L} = \prod_{i=1}^{m} \hat{y}_i^{y_i} (1 - \hat{y}_i)^{(1-y_i)}$$

Taking log:

$$\log \mathcal{L} = \sum_{i=1}^{m} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Negative log-likelihood (cost function):

$$J = -\frac{1}{m} \sum_{i=1}^{m} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

This is called the **Binary Cross-Entropy (BCE)** loss.

Derivation of Binary Cross-Entropy Loss in Logistic Regression

Logistic Regression Model

For an input vector $x \in \mathbb{R}^n$:

$$z = w^T x + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

where $\hat{y} = P(y = 1|x)$ and $y \in \{0, 1\}$.

Bernoulli Assumption

Since the output is binary, we assume:

$$y \sim \text{Bernoulli}(\hat{y})$$

The Bernoulli probability mass function is:

$$P(y|\hat{y}) = \hat{y}^{\,y}(1 - \hat{y})^{(1-y)}$$

This compactly represents:

$$P(y = 1) = \hat{y}, \quad P(y = 0) = 1 - \hat{y}$$

Likelihood Function

For m independent training samples:

$$\mathcal{L}(w, b) = \prod_{i=1}^{m} \left(\hat{y}^{(i)}\right)^{y^{(i)}} \left(1 - \hat{y}^{(i)}\right)^{(1-y^{(i)})}$$

Log-Likelihood

Taking logarithm:

$$\log \mathcal{L}(w, b) = \sum_{i=1}^{m} \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

Negative Log-Likelihood (Cost Function)

To convert maximization into minimization:

$$J(w, b) = -\sum_{i=1}^{m} \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

Averaging over m samples:

$$J(w, b) = -\frac{1}{m} \sum_{i=1}^{m} \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

Binary Cross-Entropy Form

For a single sample:

$$L(y, \hat{y}) = -\left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}) \right]$$

This is known as the **Binary Cross-Entropy (BCE)** loss.

With L_2 Regularization (Optional)

$$J(w, b) = -\frac{1}{m} \sum_{i=1}^{m} \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right] + \frac{\lambda}{2m} \sum_{j=1}^{n} w_j^2$$

Derivation of Multiclass Cross-Entropy Loss (Softmax Regression)

Model

For input $x \in \mathbb{R}^n$ and K classes:

$$z_k = w_k^T x + b_k, \quad k = 1, 2, \dots, K$$

The Softmax function gives class probabilities:

$$\hat{y}_k = P(y = k|x) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$$

Categorical Distribution Assumption

The output follows a categorical distribution:

$$P(y|\hat{y}) = \prod_{k=1}^{K} \hat{y}_k^{y_k}$$

where y_k is one-hot encoded:

$$y_k = \begin{cases} 1, & \text{if true class is } k \\ 0, & \text{otherwise} \end{cases}$$

Likelihood Function

For m independent samples:

$$\mathcal{L}(W, b) = \prod_{i=1}^{m} \prod_{k=1}^{K} \left(\hat{y}_k^{(i)}\right)^{y_k^{(i)}}$$

Log-Likelihood

$$\log \mathcal{L}(W, b) = \sum_{i=1}^{m} \sum_{k=1}^{K} y_k^{(i)} \log \hat{y}_k^{(i)}$$

Negative Log-Likelihood (Cost Function)

Minimizing the negative log-likelihood:

$$J(W, b) = -\frac{1}{m} \sum_{i=1}^{m} \sum_{k=1}^{K} y_k^{(i)} \log \hat{y}_k^{(i)}$$

This is called the **Categorical Cross-Entropy Loss**.

With L_2 Regularization (Optional)

$$J(W, b) = -\frac{1}{m} \sum_{i=1}^{m} \sum_{k=1}^{K} y_k^{(i)} \log \hat{y}_k^{(i)} + \frac{\lambda}{2m} \sum_{k=1}^{K} \|w_k\|^2$$

Multiclass Cross-Entropy Without One-Hot Encoding

For input x and K classes:

$$z_k = w_k^T x + b_k$$

Softmax probabilities:

$$\hat{y}_k = P(y = k|x) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$$

Assume the label is given as a class index:

$$y^{(i)} \in \{1, 2, \ldots, K\}$$

Likelihood

The probability of the correct class is:

$$P(y^{(i)}|x^{(i)}) = \hat{y}_{y^{(i)}}^{(i)}$$

For m independent samples:

$$\mathcal{L}(W, b) = \prod_{i=1}^{m} \hat{y}_{y^{(i)}}^{(i)}$$

Negative Log-Likelihood (Cost Function)

$$J(W, b) = -\frac{1}{m} \sum_{i=1}^{m} \log \hat{y}^{(i)}_{y^{(i)}}$$

This is the sparse categorical cross-entropy loss.